MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6J6WB 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification sheet and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	2a	1a	h	h	1g	2g	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg/g	Ik/Section	Pp/p	TE	Alt
Unit:	V	V dc	V dc	v	Ohms	Meg	mA dc	W	°C	ft
Maximum:	6.9	300	---	180	---	0.25	12.5	1.1	+165	(see note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	---	50	---	---	---	---	---

GENERAL:

Qualification - Required

1/ Formerly type 6J6WA

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	420	480	mA
1336	Heater-cathode leakage			0.4	II	Ihk	---	7	µA dc
1266	Total grid current	2,3	Eb = 250 V dc; Rk = 500 ohms; Rg = 1.0 Meg	0.4	II	Ic	0	-0.5	µA dc
1256	Electrode current (1) (anode)	2,4		0.4	II	Ib	6.5	12	mA dc
1236	Power oscillation	5	F = 250 MHz; Eb = 150 V dc; Rg = 2,000 ohms	0.4	II	Po	0.7	---	W
1306	Transconductance (1)	4		0.4	II	Sm	4,500	7,500	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	6		2.5	I	---	---	---	---
1306	Transconductance (2)	4	Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1266	Grid emission	3,7	Ef = 7.5 V; Ec = -14.5 V dc; Rg = 0.5 Meg	2.5	I	Ic	0	-1.0	µA dc
2201	Noise and microphonics	3	Ef = 6.3 V ac; Ehk = 0; Rp = 10,000 ohms; Ecal = 50 mV ac	2.5	I	---	---	---	---
1316	Amplification factor	4		6.5	S3	Mu	28	48	---
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cgp	1.0	1.6	pF
			No shield			Cin	1.4	2.8	pF
			No shield; section 2			Cout(2)	0.25	0.55	pF
			No shield; section 1			Cout(1)	0.25	0.65	pF
			No shield			Chk	4.0	8.0	pF
1031	Low-frequency vibration	3	Eb = 250 V dc; Ec = -8 V dc; Rp = 2,000 ohms; Rk = 0	6.5	Code H	Ep	---	25	mV ac
1041	Shock	8	450 G	6.5	See note 9	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1031	Vibration-fatigue			6.5	See note 9	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	Low-frequency vibration	3	Eb = 250 V dc; Ec = -8 V dc; Rp = 2,000 ohms; Rk = 0	---	---	Ep	---	50	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	20	µA dc
1306	Change in trans-conductance (1) of individual tubes	4		---	---	ΔSmt	---	20	%
1121	Base strain	10		---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1256	Electrode current (2) (anode)	4	Eb = 250 V dc; Ec = -14.5 V dc; Rk = 0	2.5	I	Ib	---	200	µA dc
1256	Electrode current (3) (anode)	4	Eb = 250 V dc; Ec = -10.5 V dc; Rk = 0	2.5	I	Ib	5.0	---	µA dc
1105	Permanence of marking			---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MIN	MAX	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = +100 V dc; Eb = 0; 1 minute "on", 4 minutes "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	15	µA dc
1516	Stability life		Ehk = +180 V dc; Rg/g = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end point (2 and 20 hours):						
1306	Change in transconductance (1) of individual tubes	4		ΔSm t	---	10	%
1501	Intermittent life	11,12	Ehk = +180 V dc; Rg/g = 0.5 Meg; TA = +165°C	---	---	---	---
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current	3	Eb = 250 V dc; Rk = 500 ohms; Rg = 1.0 Meg	Ic	0	-1.0	µA dc
1301	Heater current			If	420	480	mA dc
1306	Transconductance (1)	4		Sm	4,000	---	µmhos
1306	Transconductance (1) (average change)	4		Avg ΔSm t	---	20	%
1306	Transconductance (2)	4	Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	10	µA dc
1211	Insulation of electrodes	6		---	---	---	---

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NOTE:

1. See "reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

2. This test shall be performed at the conclusion of the holding period.

3. Tie 1g to 2g; 1a to 2a; parasitic suppressors of 50 ohms maximum shall be permitted.

4. Test each unit separately with test voltages applied to opposite section.

5. Self-excited push-pull oscillator shall be so adjusted that Ib = 33 mA dc for a bogey tube, when oscillator is tuned for maximum power output to load.

6. Test each unit separately.

7. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec	Eb	Rk	Rg
V	V dc	V dc	Ohms	Meg
7.5	0	100	50	0.5

8. A grid resistor of 0.1 megohm shall be added to each section, except when a thyratron-type short indicator is used.

9. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

10. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

    a. Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

    b. Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

11. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

12. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having boegy Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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